

Determining an Earth Observation Repeat-Ground-Track Orbit for an Optimization Methodology

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This paper presents a new methodology for a quick and efficient numerical determination of the condition for repeat ground tracks to be employed in an orbital optimization design methodology. This methodology employs the simplicity and reliability of the epicyclic motion condition for a repeat ground track to find a semimajor axis for a given repetition cycle and inclination. Then the semimajor axis is refined for application to any elliptical motion. This methodology was discovered by comparing two recent methods in addition to a new proposed method offered in this paper investigating both nonlinear algebraic and polynomial formulations of the governing repeat-ground-track condition relationship. A lesser known simplified method is used for preliminary solution refinement. The advantages and disadvantages of each approach are weighed with each method's reliability, performance, and computational ease based on a case study. From these criteria, one method is recommended for use in repeat-ground-track orbit design optimization methodology.

Nomenclature

a	= semimajor axis, km
d	= number of rotations the Earth completes during the period of repetition
e	= eccentricity
h_p	= perigee height above Earth's surface, km
i	= inclination, rad
J_2	= second-order zonal effects
J_4	= fourth-order zonal effects
k	= number of revolutions along the orbit in one period of repetition
M	= mean anomaly, rad
$\dot{M}$	= rate of change in the mean anomaly due to nominal motion and perturbations, rad/s
M_o	= initial mean anomaly, rad
N_d	= number of sidereal days the Earth completes during the period of repetition
N_p	= number of revolutions along the orbit in one period of repetition
n	= satellite's mean motion, rad/s
n_e	= epicycle frequency, rad/s
p	= semiparameter, km
R_E	= Earth radius, 6378.1363 km
T	= anomalistic period, s
T_r	= period of repetition, s
T_Ω	= nodal period of the satellite, s
$T_{\Omega G}$	= nodal period of Greenwich, s
ε	= total orbital specific energy, km ² /s ²
κ	= secular change in argument of latitude

μ	= Earth's gravitational constant, 3.986004415×10^5 km ³ /s ²
ν	= secular change in ascending node
τ	= similitude parameter
Ω	= ascending node, rad
$\dot{\Omega}$	= nodal regression of the orbital plane, rad/s
ω	= argument of perigee, rad
$\dot{\omega}$	= rate of change in the argument of perigee, rad/s
ω_E	= Earth's rotation, $7.29211585530 \times 10^{-5}$ rad/s

I. Introduction

FOR MANY Earth observation satellites, an ideal orbit designed in the presence of perturbations is based on a repeating ground track that allows for specific observations to be scheduled on a routine basis or with the same sensing conditions. The purpose of this research is to find the orbital shape (size and eccentricity) for a desired satellite inclination, height of perigee, and period of repetition for any Earth-based elliptical orbit. The orbital characteristics determined by this research are employed within a larger algorithm optimization methodology, thus the need for an efficient and accurate orbital shape solution. To determine a reliable, efficient methodology for finding the orbital shape, two recent methods for finding a repeat ground track are compared [1,2]. Unfortunately, each method is based on different assumptions and parameters, and has their own set of limitations on the type of orbit for which the method should be applied. To bridge these differences and limitations and provide a comparison for evaluating the merits and/or deficiencies, a third method is developed [3].

The first method for generating repeat ground tracks is the flower constellation (FC) concept by Mortari et al. [1]. The FC method is based on the Keplerian orbital elements: semimajor axis, eccentricity, inclination, longitude of the ascending node, argument of perigee, and the mean anomaly. The main advantage of this method is a general applicability to any elliptic eccentricity based on a specified perigee height. Unfortunately, the FC method only considers the second-order zonal effects, that is J_2 effects. The second method provides a repeating ground track based on epicyclic motion theory by Aorpimai and Palmer [2]. The epicyclic motion repeating-ground-track orbit (ERO) concept is based on two polar elements (radius and argument of latitude) and two orbital elements (inclination and longitude of the ascending node). This method was developed for near-circular orbits and therefore does not use eccentricity or require

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a specified height of perigee. The ERO method includes the J_2 term like the FC method, but also uses the J_4 first-order and J_2 second-order terms. These different perturbation models, variable sets, and orbit families make comparison between the methods difficult.

The new method is an extension or modification of the FC method, the modified flower constellation (MFC). To bridge the gap between the limitation of the FC framework using only the J_2 term and the ERO limitation for near-circular orbits, the flower constellation procedure is modified to include the J_4 and J_2^2 terms. None of these three methods provide a completely reliable, efficient, and accurate way to determine the orbital shape for any Earth-based elliptical orbit. Therefore, a fourth simplified repeat groundtrack (SRG) method by Collins [4] is applied to the preliminary FC, ERO, and MFC solutions to determine the orbital shape applicable to any Earth-based elliptical orbit. In the following sections, mathematical conditions for repeating ground tracks using the FC and ERO techniques are reviewed, a modified FC method is developed, and the SRG technique is applied as a refinement process. The modified FC method is applied to several examples and results are compared with the corresponding FC and ERO results. The FC, MFC, and ERO conditions are solved through numerical and polynomial methods for computational comparison. Finally, conclusions from these examples and results are drawn to support the selection of the refined ERO method for use in an orbit design optimization algorithm [5].

II. Condition for Repeating Ground Tracks

For an orbit to have a repeating ground track, the period of repetition T_r is related to the nodal period of the satellite T_Ω and nodal period of Greenwich [1] $T_{\Omega G}$ by Eq. (1):

$$T_r = N_p T_\Omega = N_d T_{\Omega G} \quad (1)$$

where N_p is the number of revolutions along the orbit in one period of repetition and N_d is the number of sidereal days the Earth completes during the period of repetition [1]. Note N_p and N_d are integers.

The geometry underlying this condition is illustrated in Fig. 1. The subindices i indicate the initial position and ECF is Earth-centered fixed frame. Equation (1) simply states that the time for the westward progression of the ground track to complete one revolution and the time for the northward penetration of the ground track to also complete one revolution constitute a rational number. By expressing the nodal periods to depend on the orbital elements, Eq. (1) is converted to a more useful form.

The nodal period T_Ω is related to the satellite's anomalistic period T by Eq. (2) where $\dot{M}$ denotes the rate of change in the mean anomaly due to nominal motion and perturbations and $\dot{\omega}$ is the rate of change in the argument of perigee:

$$T_\Omega = \frac{2\pi}{\dot{M} + \dot{\omega}} \quad (2)$$

The current mean anomaly, M , in Fig. 1 represents the motion of the spacecraft with respect to time. M is related to the initial mean anomaly M_o by Eq. (3) [6]:

$$M = M_o + nt \quad (3)$$

where n is the satellite's mean motion. The rate of change of the current mean anomaly is related to the rate of change of the initial mean anomaly through Eq. (4) [6]:

$$\dot{M} = \dot{M}_o + n \quad (4)$$

The anomalistic period T is similarly related to the mean motion in Eq. (5):

$$T = \frac{2\pi}{n} \quad (5)$$

Now Eq. (2) can be rewritten in terms of the anomalistic period in Eq. (6):

$$T_\Omega = \frac{2\pi}{n + \dot{M}_o + \dot{\omega}} = T \left(1 + \frac{\dot{M}_o + \dot{\omega}}{n} \right)^{-1} \quad (6)$$

Figure 1 represents the perturbing elements appearing in Eq. (6) as a function of time for the satellite.

The nodal period of Greenwich is provided as in Eq. (7):

$$T_{\Omega G} = \frac{2\pi}{\omega_E - \dot{\Omega}} \quad (7)$$

The nodal regression of the orbital plane is $\dot{\Omega}$. Referring back to Fig. 1, $T_{\Omega G}$ physically represents the time required for the Greenwich meridian to depart from the ascending node and return to its perturbed ($\dot{\Omega}$) location. Substituting these expressions for the nodal period of the satellite and nodal period of Greenwich into Eq. (1) and rearranging for the similitude parameter, $\tau = N_d/N_p$, yields Eq. (8):

$$\tau = \frac{\omega_E - \dot{\Omega}}{n + \dot{M}_o + \dot{\omega}} \quad (8)$$

Equation (8) is the fundamental necessary requirement in usable form for repeating ground tracks and is the foundation for any mission design method, including the FC and ERO methods. For a specified similitude τ , the equation will be solved for the unknown orbital parameterization variables in the various methodologies.

A. Flower Constellation

The flower constellation concept was developed by Mortari et al. [1] to group existing constellations and orbits with three common characteristics. The first characteristic is identical orbit shape. Anomalistic period, argument of perigee, height of perigee, and inclination are common across the orbit family. The second characteristic is compatible periods. The orbital period is evaluated in such a way as to yield a perfectly repeated ground track. The final characteristic is equally displaced nodes. Tracks crossing the equatorial plane at the nodes for each satellite in a complete flower constellation are displaced equally. Examples include the Walker constellation, Molniya orbits, and geosynchronous orbits. The common framework also allows for the development of new constellations with these same characteristics. A full FC may have multiple orbits, but this paper will only develop one orbit with a repeating ground track. Therefore, characteristics 1 and 3 will not be investigated in this report. The development in this report is categorized as an "incomplete" FC, since only one satellite for one orbit is addressed, not multiple orbits or a multisatellite orbit.

The FC method is based on the Keplerian orbital elements: semimajor axis, eccentricity, inclination, longitude of the ascending node, argument of perigee, and the mean anomaly. The main

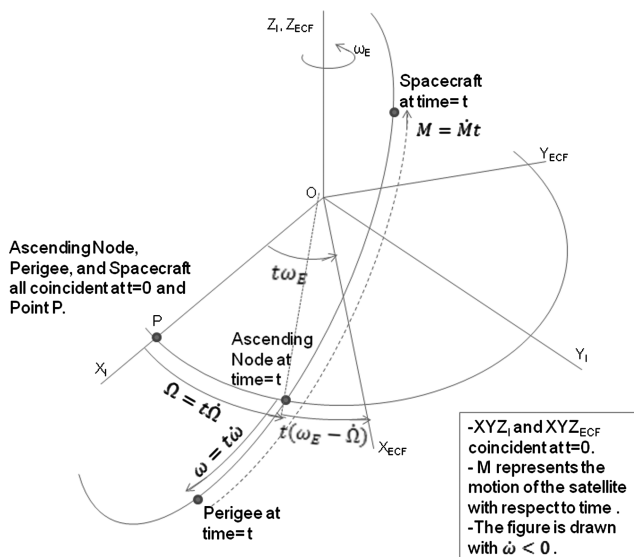


Fig. 1 Perturbing elements for repeating ground tracks [3].

advantage of this method is a general applicability for any eccentricity that is a product of a specified height of perigee. Unfortunately, the FC method only considers the second-order zonal effects, that is J_2 effects [7]. The FC concept uses geopotential perturbation theory, but the second-order zonal effects only change the argument of perigee, longitude of the ascending node, and mean anomaly [1] as applied in Eqs. (9–11):

$$\dot{\omega} = \frac{3nR_E^2 J_2}{4p^2} \{4 - 5\sin^2 i\} \quad (9)$$

$$\dot{\Omega} = -\frac{3nR_E^2 J_2}{2p^2} \cos i \quad (10)$$

$$\dot{M}_o = \frac{-3nR_E^2 J_2 \sqrt{1-e^2}}{4p^2} \{3\sin^2 i - 2\} \quad (11)$$

Equations (9–11) are derived by a standard variation of parameters technique applied to the orbital elements.

Express eccentricity, semiparameter, and mean motion in terms of the semimajor axis a and perigee height above Earth's surface h_p are given [1] in Eqs. (12–14):

$$e = 1 - \frac{R_E + h_p}{a} \quad (12)$$

$$p = 2(R_E + h_p) - \frac{(R_E + h_p)^2}{a} \quad (13)$$

$$n = \sqrt{\frac{\mu}{a^3}} \quad (14)$$

where $\mu = 3.986004415 \times 10^5 \text{ km}^3/\text{s}^2$ for the Earth's gravitational constant. Substituting Eqs. (9–14) into the similitude expression yields Eq. (15) [3]:

$$\tau = \frac{\omega_E + 2A(a) \cos i}{\sqrt{\frac{\mu}{a^3}} + A(a) \left[\sqrt{1 - \left(1 - \frac{R_E + h_p}{a}\right)^2} \{2 - 3\sin^2 i\} + \{4 - 5\sin^2 i\} \right]} \quad (15)$$

where

$$A(a) = \frac{3\sqrt{\frac{\mu}{a^3}} R_E^2 J_2}{4 \left(2(R_E + h_p) - \frac{(R_E + h_p)^2}{a} \right)^2}$$

Equation (15) will be used to compute the semimajor axis for specific inclination and perigee height that result in a repeat ground track. For computational advantages, Eq. (15) can be converted to the polynomial equation given in Eq. (16) [3]:

$$4\mu\phi a^2 y(a)^2 x(a) = [\omega_E^2 a^3 x(a)^4 - \phi a^2 x(a) - \mu y(a)^2] \quad (16)$$

where

$$\begin{aligned} \beta &= 2 - 3\sin^2 i & \delta &= \frac{3R_E^2 J_2}{4} \\ \gamma &= 4 - 5\sin^2 i & \phi &= \mu \tau^2 \delta^2 \beta^2 & x(a) &= ap \\ y(a) &= \tau [x(a)^2 + \delta \gamma a^2] - 2\delta a^2 \cos i \end{aligned}$$

The semimajor axis can be determined from Eq. (16) for a given inclination and specified height of perigee.

B. Epicyclic Motion Repeat-Ground-Track Orbit

The second method provides a repeating ground track based on epicyclic motion theory by Aorpimai and Palmer [2]. The ERO methodology is based on two polar elements (radius and argument of latitude) and two orbital elements (inclination and longitude of the ascending node). The argument of latitude replaces the argument of perigee and mean anomaly. ERO includes the J_2 term like the FC, but also uses the J_4 first-order and J_2 second-order terms. This method was developed for near-circular orbits with an eccentricity less than the order of J_2 and therefore does not use eccentricity or require a specified height of perigee [8].

The ERO basis has a very similar development of perturbation theory to the FC method but uses the argument of latitude instead of the argument of perigee and mean anomaly [2]. This choice is consistent with the near-circular assumption and omits eccentricity. The ERO development is different from other published techniques because only a mean orbital radius is defined. A mean orbital plane or other mean elements are not defined in the ERO technique [9]. The ERO semimajor axis a is designed to satisfy the energy relation equation Eq. (17):

$$\varepsilon = -\left(\frac{\mu}{2a}\right) \quad (17)$$

ε is the total orbital specific energy for the satellite and constant for an axisymmetric gravitational potential. The semimajor axis for a circular orbit and epicycle frequency, n_e , satisfy $a^3 n_e^2 = \mu$. The nodal frequency used in the FC method is related to the epicycle anomalistic frequency through the relation seen in Eq. (18) [9]:

$$\frac{2\pi}{T_\Omega} = (1 + \kappa)n_e \quad (18)$$

where κ is the secular change in argument of latitude.

Equation (19) provides the basic ERO repeating-ground-track condition equivalent to Eq. (8):

$$\frac{d}{k} = \frac{\frac{\omega_E}{n_e} - \nu}{1 + \kappa} \quad (19)$$

Parameters d and k are analogous to FC parameters N_d and N_p , and ν is the secular change in ascending node. Using the defined terms of ν and κ , [9], where

$$\begin{aligned} \kappa &= \frac{3}{4} J_2 \left(\frac{R_E}{a}\right)^2 (4 - 5\sin^2 i) - \frac{3}{64} J_4 \left(\frac{R_E}{a}\right)^4 (136 - 500\sin^2 i \\ &\quad + 385\sin^4 i) + \frac{3}{16} J_2^2 \left(\frac{R_E}{a}\right)^4 (14 + 17\sin^2 i - 35\sin^4 i) \end{aligned} \quad (20)$$

$$\begin{aligned} \nu &= -\frac{3}{2} J_2 \left(\frac{R_E}{a}\right)^2 \cos i + \frac{15}{16} J_4 \left(\frac{R_E}{a}\right)^4 \cos i (4 - 7\sin^2 i) \\ &\quad + \frac{3}{8} J_2^2 \left(\frac{R_E}{a}\right)^4 \cos i (11 - 20\sin^2 i) \end{aligned} \quad (21)$$

the repeat condition can be written in terms of semimajor axis and inclination [2] given in Eq. (22) using J_2 , J_4 , and J_2^2 perturbations:

$$\frac{d}{k} = \frac{\frac{\omega_E}{J_2} \sqrt{\frac{a^3}{\mu}} \left(\frac{R_E}{a}\right)^2 + \frac{3}{2} \cos i - \frac{15}{16} \frac{J_4}{J_2} \left(\frac{R_E}{a}\right)^2 \cos i (4 - 7\sin^2 i) - \frac{3}{8} \frac{J_2^2}{J_2} \left(\frac{R_E}{a}\right)^2 \cos i (11 - 20\sin^2 i)}{\left(\frac{a}{R_E}\right)^2 \frac{1}{J_2} + \left(3 - \frac{15}{4} \sin^2 i\right) - \frac{15}{16} \frac{J_4}{J_2} \left(\frac{R_E}{a}\right)^2 \left(\frac{34}{5} - 25\sin^2 i + \frac{77}{4} \sin^4 i\right) + \frac{3}{16} \frac{J_2^2}{J_2} \left(\frac{R_E}{a}\right)^2 (14 + 17\sin^2 i - 35\sin^4 i)} \quad (22)$$

Equation (22) can be used to determine the semimajor axis for a given inclination using a numerical solver.

For computational advantages, Eq. (22) can be converted to the polynomial equation given in Eq. (23) [3]:

$$\{\tau(\zeta a^4 + \eta a^2 - \theta + \lambda) - \gamma a^2 + \delta + \varepsilon\}^2 - \beta^2 a^{11} = 0 \quad (23)$$

where

$$\begin{aligned} \beta &= \frac{\omega_E}{J_2 R_E^2 \sqrt{\mu}} & \gamma &= \frac{3}{2} \cos i & \delta &= \frac{15 J_4}{16 J_2} R_E^2 \cos i \{4 - 7 \sin^2 i\} \\ \varepsilon &= \frac{3 J_2^2}{8 J_2} R_E^2 \cos i \{11 - 20 \sin^2 i\} & \zeta &= \frac{1}{J_2 R_E^2} \\ \eta &= 3 - \frac{15}{4} \sin^2 i & \theta &= \frac{15 J_4}{16 J_2} R_E^2 \left\{ \frac{34}{5} - 25 \sin^2 i + \frac{77}{4} \sin^4 i \right\} \\ \lambda &= \frac{3 J_2^2}{16 J_2} R_E^2 \{14 + 17 \sin^2 i - 35 \sin^4 i\} \end{aligned}$$

C. Modified Flower Constellation

The modified flower constellation (MFC) uses a similar development as the basic flower constellation approach, except the MFC method includes additional gravitational terms in the variations in argument of perigee, longitude of the ascending node, and mean anomaly. This enhancement allows for additional perturbation terms in order to compare with the ERO while retaining eccentricity and the ability to address noncircular orbits [3]. Adding the J_4 and J_2^2 terms [6] yields the following variational expressions in Eqs. (24) and (26):

$$\begin{aligned} \dot{\omega} &= \frac{3 J_2 R_E^2}{4 p^2} \{4 - 5 \sin^2 i\} + \frac{9 n J_2^2 R_E^4}{384 p^4} \{56 e^2 + (760 - 36 e^2) \sin^2 i \\ &\quad - (890 + 45 e^2) \sin^4 i\} - \frac{15 n J_4 R_E^4}{128 p^4} \{64 + 72 e^2 \\ &\quad - (248 + 252 e^2) \sin^2 i + (196 + 189 e^2) \sin^4 i\} \end{aligned} \quad (24)$$

$$\begin{aligned} \dot{\Omega} &= \frac{-3 J_2 R_E^2 n \cos i}{2 p^2} + \frac{3 J_2^2 R_E^4 n \cos i}{32 p^4} \{12 - 4 e^2 - (80 + 5 e^2) \sin^2 i\} \\ &\quad + \frac{15 J_4 R_E^4 n \cos i}{32 p^4} \{8 + 12 e^2 - (14 + 21 e^2) \sin^2 i\} \end{aligned} \quad (25)$$

$$\begin{aligned} \dot{M}_o &= \frac{3 J_2 R_E^2 n \sqrt{1 - e^2}}{4 p^2} \{2 - 3 \sin^2 i\} + \frac{3 n J_2^2 R_E^4}{512 p^4 \sqrt{1 - e^2}} \\ &\quad \times \{320 e^2 - 280 e^4 + (1600 - 1568 e^2 + 328 e^4) \sin^2 i \\ &\quad + (-2096 + 1072 e^2 + 79 e^4) \sin^4 i\} \\ &\quad - \frac{45 J_4 R_E^4 e^2 n \sqrt{1 - e^2}}{128 p^4} \{-8 + 40 \sin i - 35 \sin^2 i\} \end{aligned} \quad (26)$$

Once again substitute Eqs. (24–26) into the basic condition for a repeating ground track in Eq. (8). Also replace e , p , and n as given in Eqs. (12–14) to yield the equation that is used to determine the semimajor axis for a given inclination and perigee height. The MFC formulation preserves eccentricity like the FC and includes the additional J_4 and J_2^2 terms like the ERO. The MFC condition is given in Eq. (27):

where

$$\begin{aligned} x &= x(a) = ap & \delta &= \frac{3 R_E^2 J_2}{4} & \beta &= \frac{3 R_E^4 J_2^2}{1536} \\ \gamma &= \frac{15 R_E^4 J_4}{128} & \varepsilon &= b\beta + f\gamma + d\beta + g\gamma & \zeta &= d\beta + g\gamma \\ \eta &= v\beta + t\gamma + w\beta + u\gamma & \phi &= w\beta + u\gamma \\ \lambda &= (q + k + m)\beta & v &= (2q + m)\beta \\ b &= 48 \cos i (12 - 80 \sin^2 i) & d &= 48 \cos i (-4 - 5 \sin^2 i) \\ f &= 4 \cos i (8 - 14 \sin^2 i) & g &= 4 \cos i (12 - 21 \sin^2 i) \\ h &= 2 - 3 \sin^2 i & j &= 4 - 5 \sin^2 i & k &= 3(1600 \sin^2 i - 2096 \sin^4 i) \\ m &= 3(320 - 1568 \sin^2 i + 1072 \sin^4 i) \\ q &= 3(-280 + 328 \sin^2 i + 79 \sin^4 i) & s &= 3(8 - 40 \sin i + 35 \sin^2 i) \\ t &= -(64 - 248 \sin^2 i + 196 \sin^4 i) \\ u &= -(72 - 252 \sin^2 i + 189 \sin^4 i) \\ v &= 12(760 \sin^2 i - 890 \sin^4 i) & w &= 12(56 - 36 \sin^2 i - 45 \sin^4 i) \end{aligned}$$

The MFC condition can also be transformed into a polynomial equation given in Eq. (28) [3]:

$$\begin{aligned} \{\tau^2 \mu E(a)^2 - \omega_E^2 a^3 x(a)^9 - \mu x(a)[D(a) - \tau F(a)]^2\}^2 \\ - 4 \omega_E^2 \mu a^3 x(a)^{10} [D(a) - \tau F(a)]^2 = 0 \end{aligned} \quad (28)$$

where

$$\begin{aligned} x(a) &= ap & D(a) &= 2 \cos(i) \delta a^2 x^2 - \varepsilon a^4 + \zeta a^2 x \\ E(a) &= h \delta a x^3 + s \gamma a^3 x - s \gamma a x^2 + \lambda a^5 - v a^3 x + q \beta a x^2 \\ F(a) &= x^4 + j \delta a^2 x^2 + \eta a^4 - \phi a^2 x & \delta &= \frac{3 R_E^2 J_2}{4} \\ \beta &= \frac{3 R_E^4 J_2^2}{1536} & \gamma &= \frac{15 R_E^4 J_4}{128} & \varepsilon &= b\beta + f\gamma + d\beta + g\gamma \\ \zeta &= d\beta + g\gamma & \eta &= v\beta + t\gamma + w\beta + u\gamma & \phi &= w\beta + u\gamma \\ \lambda &= (q + k + m)\beta & v &= (2q + m)\beta \\ b &= 48 \cos i (12 - 80 \sin^2 i) & d &= 48 \cos i (-4 - 5 \sin^2 i) \\ f &= 4 \cos i (8 - 14 \sin^2 i) & g &= 4 \cos i (12 - 21 \sin^2 i) \\ h &= 2 - 3 \sin^2 i & j &= 4 - 5 \sin^2 i & k &= 3(1600 \sin^2 i - 2096 \sin^4 i) \\ m &= 3(320 - 1568 \sin^2 i + 1072 \sin^4 i) \\ q &= 3(-280 + 328 \sin^2 i + 79 \sin^4 i) \\ s &= 3(8 - 40 \sin i + 35 \sin^2 i) & t &= -(64 - 248 \sin^2 i + 196 \sin^4 i) \\ u &= -(72 - 252 \sin^2 i + 189 \sin^4 i) & v &= 12(760 \sin^2 i - 890 \sin^4 i) \\ w &= 12(56 - 36 \sin^2 i - 45 \sin^4 i) \end{aligned}$$

III. Simplified Repeat Groundtrack

Another method of evaluating the repeat-ground-track method is to use a simplified form of Eq. (8) with only the J_2 term [4]. This method is different from the FC method in that eccentricity is given,

$$\tau = \frac{\omega_E + \sqrt{\frac{\mu}{a}} \left(2 \cos(i) \delta \frac{a}{x^2} - \varepsilon \frac{a^3}{x^4} + \zeta \frac{a}{x^2} \right)}{\sqrt{\frac{\mu}{a}} \left[\frac{1}{a} + h \delta \frac{\sqrt{x}}{x^2} + s \gamma \frac{\sqrt{x}}{x^3} \left(\frac{a^2}{x} - 1 \right) + j \delta \frac{a}{x^2} + \eta \frac{a^3}{x^4} - \phi \frac{a}{x^3} + \lambda \frac{a^4}{x^5 \sqrt{x}} - v \frac{a^2}{x^3 \sqrt{x}} + q \beta \frac{1}{x^2 \sqrt{x}} \right]} \quad (29)$$

not a function of the perigee height and semimajor axis. Equation (8) may be written as

$$n \left(1 + \frac{3}{2} J_2 \frac{R_E^2}{a^2} \psi \right) - \frac{\omega_E}{\tau} = 0 \quad (29)$$

where

$$\psi = 3(1 - e^2)^{-\frac{3}{2}} \left(\frac{1}{3} - \frac{1}{2} \sin^2 i \right) - \frac{1}{(1 - e^2)^2} \left(\frac{1}{2} - \frac{5}{2} \cos^2 i \right) - \frac{1/\tau}{(1 - e^2)^2} \cos i$$

Let

$$Q = \frac{3}{2} J_2 \frac{R_E^2}{\mu^{2/3}} \psi \quad (30)$$

and

$$\chi = n^{1/3} \quad (31)$$

so that Eq. (29) becomes

$$\chi^3 + Q\chi^7 - \frac{\omega_E}{\tau} = 0 \quad (32)$$

Equation (32) can be easily solved using a two-step iteration Newton method with a good initial guess for the semimajor axis and given inclination, eccentricity, and similitude parameter.[‡]

The SRG refinement method is needed because the FC and MFC methods make the assumption $n = (\mu/a^3)^{1/2}$, which is not true under perturbation and introduces error in the computed semimajor axis. Although simulating the perturbed orbital motion with such values leads to a ground trace that is approximately closed, the SRG solution provides a higher accuracy in ground-track closure. The refinement method is used on the ERO method to provide for any eccentric orbital shape and thereby solving for a semimajor axis in equivalent terms as the refined FC and MFC, changing from the original ERO epicyclic semimajor-axis solution. For near-circular solutions, the original ERO solution provides a sufficient solution [2]; however, the intent of this paper is to compare methods for any elliptical shape orbit. Given a starting condition for a from either the FC, ERO, or MFC methods, Eq. (32) is used to find $\chi(n)$ directly and $n = (\mu/a^3)^{1/2}$ is used to find the corresponding a , leading to the two-step iteration procedure. Usually, only two iterations are needed for convergence.

IV. Case Study

The case study focuses on evaluating the four processes to discover the most efficient method for finding a semimajor axis for a given inclination, similitude parameter, and perigee height. The equations for repeating-ground-track conditions for FC, ERO, MFC, and SRG are the basis for comparing the four methods. The FC, ERO, and MFC equations are first used to find the semimajor axis for a given inclination, similitude parameter, and perigee height. The SRG method needs a close guess initial semimajor axis, as well as given eccentricity, inclination, and similitude parameter. Thus, the case study uses the SRG method to refine the semimajor axis determined by the FC, ERO, and MFC methods called FCr, EROr, and MFCr, respectively.

A grid pattern is laid with ranging inclinations and eccentricity for three cases for $\tau = \frac{1}{2}$ (12 h period), $\tau = \frac{1}{4}$ (6 h period), and $\tau = 1/16$ (90 min period). For $\tau = \frac{1}{2}$, e ranges from 0 to 0.68. For $\tau = \frac{1}{4}$, e ranges from 0 to 0.28. For $\tau = 1/16$, $e \approx 0$. Inclination ranges from 0 to 100 deg at a step interval of 5 deg for all three cases. Figures 2–10 show the results for the inclination versus semimajor axis for each of the FC, ERO, and MFC methods in comparison to the SRG refinement. In all three cases the SRG refinement for each method converges to the same result, and the unrefined MFC method closely

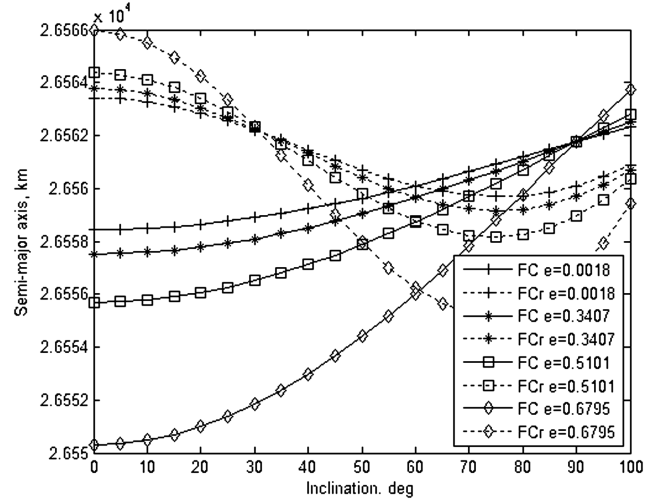


Fig. 2 Inclination vs semimajor axis for FC methods $\tau = \frac{1}{2}$.

approximates this result. This behavior shows that using the FC or ERO method with the SRG refinement leads to the same semimajor axis as determined by the MFC method without the need for refinement. Finally, ground tracks for $\tau = \frac{1}{4}$ are shown demonstrating the accuracy of ground-track closure for each method. A few additional comments are given on Figs. 2–10 before moving on. For

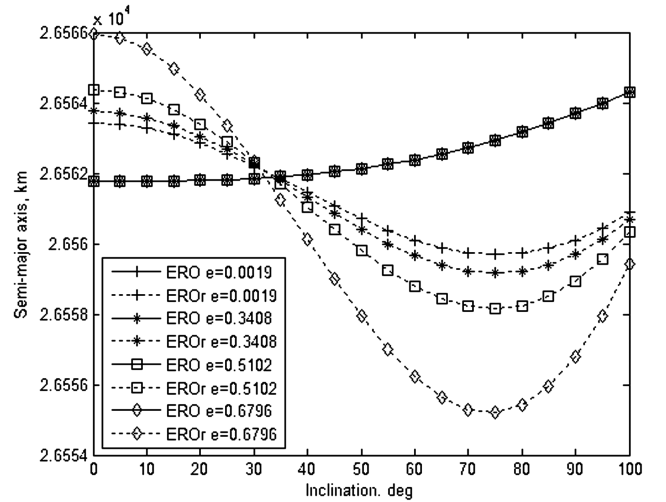


Fig. 3 Inclination vs semimajor axis for ERO methods $\tau = \frac{1}{2}$.

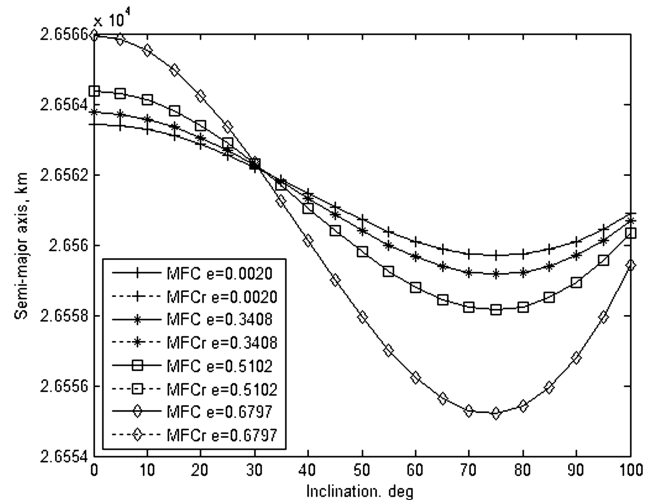


Fig. 4 Inclination vs semimajor axis for MFC methods $\tau = \frac{1}{2}$.

[‡]Private communication, P. Cefola, September 2009.

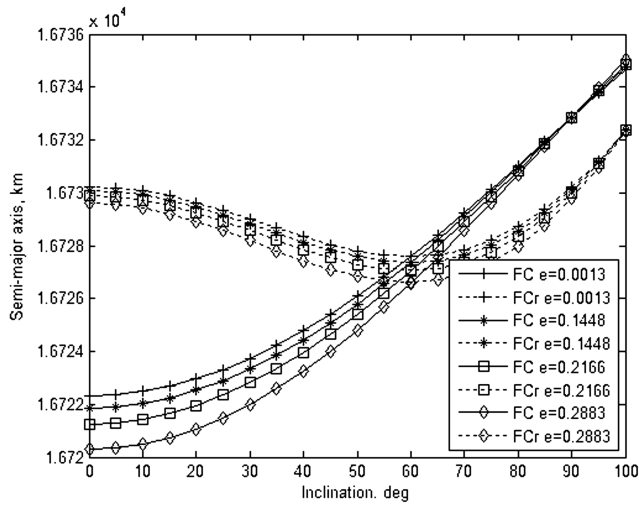


Fig. 5 Inclination vs semimajor axis for FC methods $\tau = \frac{1}{4}$.

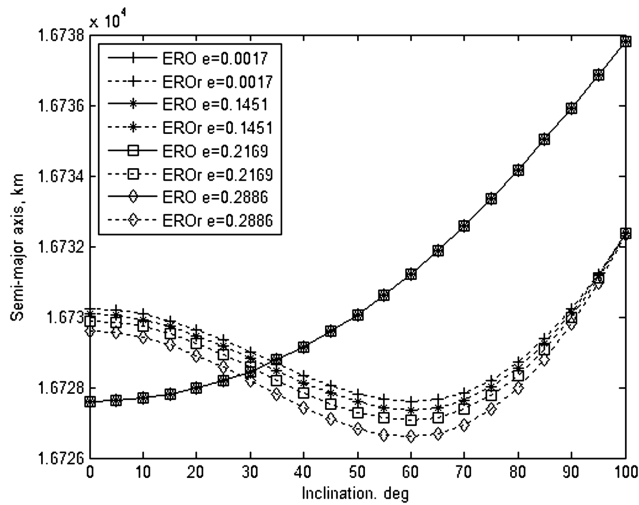


Fig. 6 Inclination vs semimajor axis for ERO methods $\tau = \frac{1}{4}$.

the $\tau = \frac{1}{2}$ case, note that all FC semimajor-axis curves over the family of eccentricities coalesce at the inclination of 90 deg. This behavior is likely related to the $\cos i$ numerator term in Eq. (15) vanishing at this critical inclination. Also note the unrefined and refined curves for each eccentricity intersect precisely at 60 deg inclination. For this

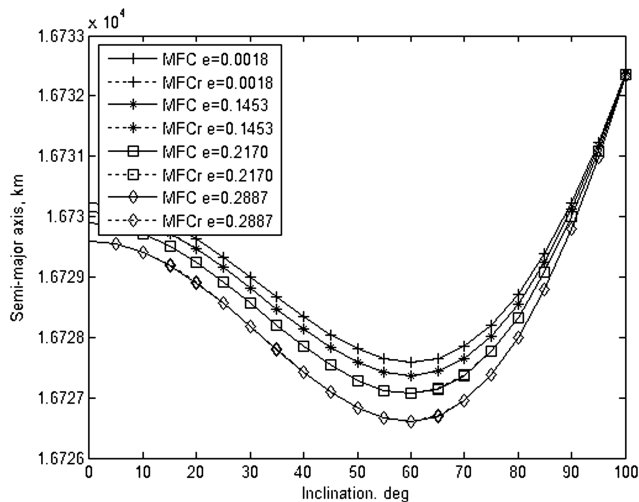


Fig. 7 Inclination vs semimajor axis for MFC methods $\tau = \frac{1}{4}$.

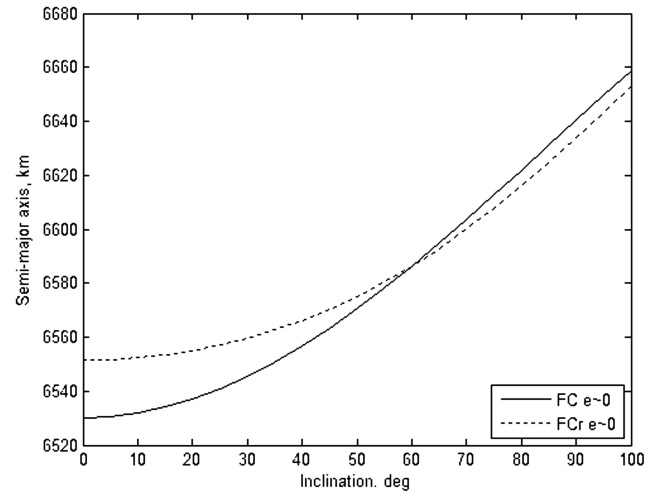


Fig. 8 Inclination vs semimajor axis for FC methods $\tau = 1/16$.

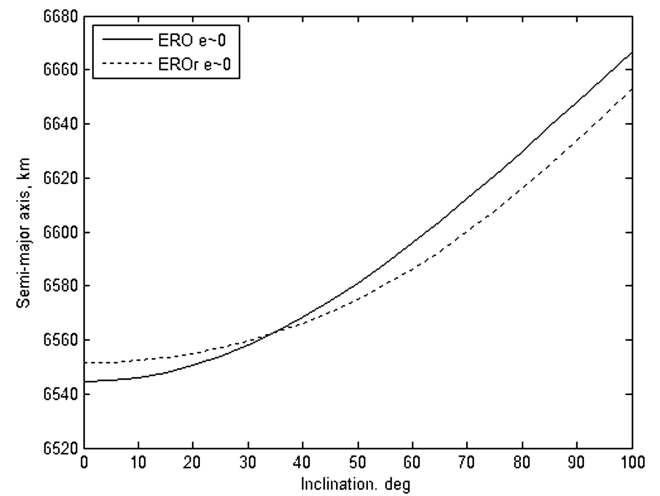


Fig. 9 Inclination vs semimajor axis for ERO methods $\tau = 1/16$.

value of i , each $\sin^2 i$ factor in the denominator in Eq. (15) equates to ± 0.25 , while in Eq. (29) both $\sin^2 i$ and $\cos^2 i$ factors within parameter ψ become ± 0.125 (including multiplying the term 3) at $i = 60$ deg. The noted intersections in Fig. 2 correlate with this term canceling. Similar features are observed in the FC method for $\tau = \frac{1}{4}$

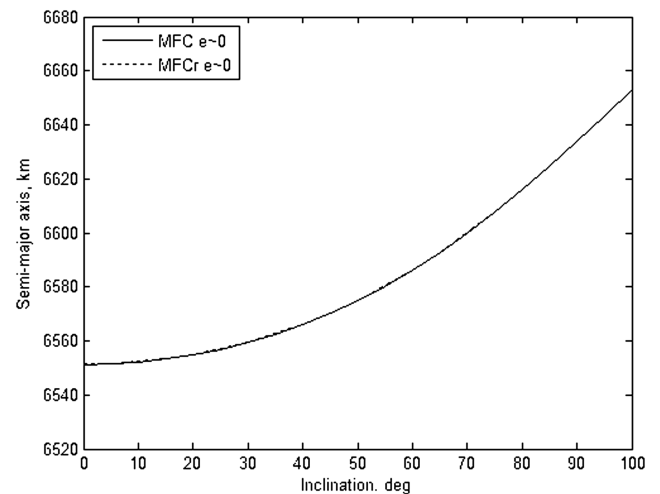


Fig. 10 Inclination vs semimajor axis for MFC methods $\tau = 1/16$.

Table 1 Magnitude of computational times

Solution method	$\tau = \frac{1}{2}$, s	$\tau = \frac{1}{4}$, s	$\tau = 1/16$, s
FC newton	10^{-3}	10^{-3}	10^{-3}
FC polynomial	10^{-4}	10^{-4}	10^{-4}
ERO newton	10^{-3}	10^{-3}	10^{-2}
ERO polynomial	10^{-4}	10^{-4}	10^{-4}
MFC newton	10^{-3}	10^{-3}	10^{-2}
MFC polynomial	10^{-3}	10^{-3}	10^{-3}

Table 2 Error function value

Solution method	$\tau = \frac{1}{2}$	$\tau = \frac{1}{4}$	$\tau = 1/16$
FC	10^{-3}	10^{-3}	10^{-3}
ERO	10^{-7}	10^{-8}	10^{-8}
EROr	10^{-16}	10^{-16}	10^{-13}
MFC	10^{-8}	10^{-8}	10^{-8}

and $1/16$ in Figs. 5–8. Finally, note that the refined curves for all three methods (Figs. 2–4) intersect near $i = 30$ deg for $\tau = \frac{1}{2}$ but there is no similar intersection for $\tau = \frac{1}{4}$ and $1/16$ (Figs. 5–10). For $\tau = \frac{1}{2}$ and $i = 30$ deg, the $\sin^2 i$ factor in Eq. (29) becomes $+0.625$, and the combined $\cos^2 i$ and $\cos i$ factor gives $-1/0.622$, again causing an approximate cancellation of terms (i.e., elimination of the eccentricity parameter) thus inducing the intersection of curves. When $\tau \neq \frac{1}{2}$, this alignment is diffused.

A second part of the case study focused on the computational time needed to find the semimajor axis for a given inclination, similitude parameter, and perigee height. Once again, the ERO equation is based only on inclination and similitude parameter (near-circular orbit). The same three cases for $\tau = \frac{1}{2}$ (12 h period), $\tau = \frac{1}{4}$ (6 h period), and $\tau = 1/16$ (90 min period) are also used in comparing the computational time. The first computational time is found through using an algebraic numerical solver, the Newton method for this case study, for the semimajor axis using Eqs. (15), (22), and (27). A realistic choice for a first guess of the semimajor axis leads to an appropriate root. The second process leading to a second computational time is from the corresponding polynomial equations. The polynomial roots are found with the MATLAB `roots` command. Here all roots must be examined to determine the appropriate root. The FC and MFC polynomial equations can lead to multiple roots within a valid orbital range (from low orbital altitudes to a geosynchronous altitude.) Therefore the valid roots must be further examined to find the appropriate root. Since finding the valid roots for the FC and MFC methods is complex, the SRG refinement with a two-step iteration process is only applied to the ERO method. Table 1 shows the results for magnitude of computational time for the three processes for each τ based on the FC, ERO, and MFC approaches.

Computational times are found using the `tíc` and `tóc` commands in MATLAB. Each run represents an inclination and eccentricity pair, and a total of 105 pairs for $\tau = \frac{1}{2}$ and $\div$ were considered, but only 63 pairs were used in the $\tau = 1/16$ case.

The polynomial root-finding approach consistently outperformed the algebraic Newton iteration approach with regards to computational time. A single run using the polynomial equation formulation consistently required 0.001 s or less to complete, while the algebraic equation formulation often took 3 to 5 times longer. When using the algebraic Newton iteration approach, due to the larger number of terms to evaluate (J_4, J_2^2), the ERO and MFC runs approximately took 0.003 to 0.005 s, while the simpler FC runs usually took 0.002 s of computational time.

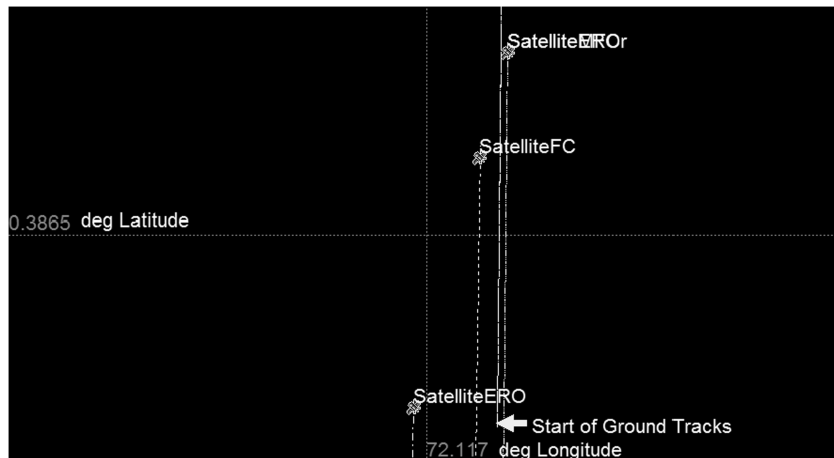
Another consideration for evaluating repeat-ground-track methods is the value of the repeat-ground-track function, f , where by rearranging Eq. (8) for the residual, or error function value:

$$f = \tau - \frac{\omega_E - \dot{\Omega}}{n + \dot{M}_o + \dot{\omega}} \quad (33)$$

For an ideal case, f is zero. For this case study, each method was evaluated using the appropriate form of Eq. (33). The tolerance selected for the Newton method was 10^{-3} in order to maintain quick computing times. Tighter tolerances will produce smaller residual error with undesirable impact on run time. The same three cases for $\tau = \frac{1}{2}$ (12 h period), $\tau = \frac{1}{4}$ (6 h period), and $\tau = 1/16$ (90 min period) are presented in comparing the function value. Also, identical gridding for inclination and eccentricity as that used in the semimajor-axis comparison was employed here.

Table 2 shows the results for the error function value for the FC, ERO, EROr, and MFC methods based on Eq. (33). Note the magnitude difference in the values. The EROr method consistently has the smallest error (10^{-13} to 10^{-16}) and is likely related to it being a refinement process of a previous solution. The ERO and MFC methods have comparable error (10^{-7} to 10^{-8}) and consistently outperform the FC method (10^{-3}). These differences in convergence between the FC and ERO/MFC methods are likely due to the differing function slopes at the semimajor-axis zero due to the various J_4, J_2^2 perturbation terms present in the ERO/MFC methods and absent in the FC method.

Figure 11 shows the accuracy of the ground-track closure for the four methods. Each method used the first eccentricity result at an inclination of 75 deg provided from Figs. 5–7. This verification of ground-track closure was run in Satellite Toolkit with the J_4 perturbation model. The zoomed-in view shows four lines, with the first line being the ERO ground track. The second line is the FC ground track. The third line is the beginning of all the ground tracks. The last line shows the agreement of the ERO refined and MFC methods. Differences in the ground-track closure are apparent in this figure. The ERO method is 15.5 km from closing, the FC method is

**Fig. 11** Ground-track closure.

3.8 km from closing, and the MFC and EROr methods are only 1.4 km from closing.

V. Conclusions

This research has shown how to quickly determine accurate orbital shape characteristics for a repeat ground track for a given set of inclination, height of perigee, and period of repetition for employment in an optimization methodology. The research first uses an epicyclic motion repeat-ground-track polynomial equation based on inclination and repeat timing to find a semimajor-axis solution. The epicyclic motion polynomial equation has a single viable root, which is calculated much faster than a numerical solver. The epicyclic motion semimajor-axis solution is then refined to include eccentricity and therefore be applicable to any elliptical motion with high precision ground-track closure. This method provides a simple, effective algorithm to determine the semimajor axis for a repeat-ground-track orbit.

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